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Performance improvement of phase change materials encapsulated with graphene oxide for thermal storage

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ABSTRACT

The thermal enhanced phase change microcapsules were prepared by paraffin as core material and the modified graphene as wall material. X-ray diffraction (XRD) and Fourier transform infrared spectroscope (FTIR) were applied to determine the microstructure and chemical structure of the microcapsules. A differential scanning calorimeter (DSC) test was done to investigate the thermal properties and thermal stability of the microcapsules. The results showed that the modified graphene/paraffin microcapsules have a regular spherical shape. The thermal conductivity of the microcapsules increases gradually with the increase of the content of the modified graphene. When the content of graphene content is 10 %, the thermal conductivity of the microcapsules is 4.01 times that of paraffin. The enthalpy of phase transition of the microcapsules is 100.3 J/g. Considering the suitable thermal properties, good thermal reliability, chemical stability and great thermal conductivity of the prepared microcapsules, it shows good application prospects in thermal or solar energy storage applications.

1. Introduction

Phase change materials (PCM) can absorb or release heat according to the change of ambient temperature so as to achieve the purpose of regulating temperature and saving energy [1,2]. PCMs have been widely used in construction, solar energy storage, medicine, agriculture and other fields. However, in the process of heating and cooling, the volume of most PCMs will change and the liquid PCM is prone to leak. Besides, its corrosiveness and low thermal conductivity are also its disadvantages. These problems hinder its application. The key to solving these problems is to package PCMs with some encapsulation materials. Microcapsule technology, using polymer wall materials to wrap up phase change materials to prepare phase change microcapsules, is the most widely used encapsulation technology. However, due to the low thermal conductivity of the polymer wall material, the thermal response of the phase change microcapsules with polymer wall is low.

How to improve the thermal conductivity of PCMs is still a key point in the field of phase change energy storage fields. Now research is focusing on filling metals or compounds that have high thermal conductivity in PCMs. Chen Ling et al. [3] used melamine resin and paraffin as the wall materials and core materials respectively to prepare phase change microcapsules by in-situ method. TGA test showed that, after the wall materials were inlaid nano-alumina, the thermal stability of the phase change microcapsules was improved obviously and the decomposition temperature increased by 20 °C on average. J. L. Zeng et al. [4] prepared tetradecyl/silver nanowires composite PCMs by adding silver nanowires into tetradecyl and tested the thermal stability of the composite PCMs. The results show that the thermal conductivity of the composites increases by 5 times when adding 12 vol% silver nanowires. Claudia Naldi et al. loaded PCMs with high-porosity open-cell copper and aluminum foams and proposed a new method for the calculation of the effective thermal conductivity of the PCM-medium [5]. Xiao et al. [6] prepared a novel PEG/BPC-Ag(Ag@CPCM) with navel orange peel-based porous carbon (BPC) as carrier, nano-silver as surface modification material and polyethylene glycol (PEG) as phase change medium. The results showed that the thermal conductivity of CPCMs based on umbilical cord peel was increased from 0.313 W/m K to 0.639 W/m K with the addition of nano silver. Liu et al. [7] prepared a novel microencapsulated phase change material (MicroEPCM) based on n- eicosane core and nano-silicon carbide phenolic resin shell modification by in-situ polymerization. Compared with MicroEPCM without nano additives, the thermal conductivity was increased by 61 % and 97 %, respectively. When the microcapsules are filled with the filler with a high thermal conductivity, the filling ratio is generally large in order to achieve the desired thermal conductivity, resulting in an increase in the material cost. In addition, the metal oxide is easily deposited because of

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the high density, which leads to uneven thermal conductivity of the microcapsules.

There is also some research on using carbon materials to enhance the thermal conductivity of phase change materials, such as graphene, carbon nanotubes, carbon nanofibers and so on. Zitao Yu et al. [8] studied the effect of adding carbon nano-admixtures (such as short multi-walled carbon nanotubes, long multi-walled carbon nanotubes, carbon nanofibers and graphene nanosheets) into liquid paraffin based suspensions on the thermal conductivity. The results showed that the thermal conductivity of the suspensions with graphene nanosheets (GNPs) increases significantly compared with other carbon nano-additives. Jinglei Xiang and Lawrence T. Drzal [9] studied the effects of different proportions of exfoliated graphene nanosheets on the thermal conductivity and latent heat of paraffin PCM. The results showed that nanofillers should have a large aspect ratio, a single directionality and a low interfacial density in order to obtain high thermal conductivity PCMs. Exfoliated graphene nanosheets have little effect on the latent heat of paraffin PCMs and are helpful to improve the thermal stability. Luo et al. [10] prepared PVA reduced graphene oxide (rGO) aerogels, which is very suitable for containing polyethylene glycol, and the subsequent composite PCM (PEG/PVA-rGO) shows high thermal conductivity. The thermal conductivity is also increased to 0.348 W/(MK), and the melting enthalpy of PEG/PVA-r graphene oxide is 155.5 J/g. Hasbi et al. [11] prepared BPCM such as palm wax, beeswax and coconut oil with different weight concentrations of graphene and titanium oxide. The thermal conductivity of both fillers increased between 4 and 46 % compared to pure BPCMs.

In addition, using graphene oxide to encapsulate microcapsules for the second time can enhance its thermal conductivity. Y. Zhang et al. [12] prepared double-shell phase change microcapsules by microemulsion method. They used n-hexadecane as the core material, polystyrene as the inner shell and graphene oxide as the external shell. The morphology of the prepared microcapsule is complete and the coating rate reaches 78.5 %. The microcapsules have higher thermal stability than those microcapsules without graphene oxide. Kunjie Yuan et al. [13] prepared microcapsules by coating paraffin with silica, and then they wrapped a layer of graphene oxide on the outer layer. The results show that the thermal conductivity of the phase change microcapsules rise to 1.162 W/ (m·K) after being coated with graphene oxide on the outer layer, which is 12 % higher than those without coating. Reji Kumar et al. [14] aim to address the thermal conductivity by dispersing functionalized multi-walled carbon nanotube (FMWCNT) particles into the salt hydrate PCM. The results revealed that maximum percentage improvements in thermal conductivity was found to be 100 %.

In previous studies, the thermal conductivity has been increased by filling or re-coating with carbon materials. However, graphene, carbon nanotubes etc. are hydrophobic. They are hard to disperse in the solvent to compose with other materials. Especially, well-structured graphene has good stability. It is prone to agglomeration [15,16] which causes a problem of uneven heat transfer. In addition, when graphene oxide is used to coat phase change microcapsules, graphene oxide does not form a stable chemical connection with the original microcapsule wall material, resulting in the secondary encapsulated microcapsules being easily washed and peeled off in aqueous solution. To improve the above problems, ethylenediamine graphene oxide was simultaneously reduced and modified to prepare the modified graphene (*E*-rGO) in this paper. E-rGO was utilized to encapsulates PCMs and it does not need to be dispersed inside PCMs. This method will improve the leakage of PCMs and reduce the aggregation phenomenon. Also, the thermal conductivity of PCMs will be enhanced.

2. Experimental

2.1. Experimental materials

Natural Flake Graphite (NFG) was supplied from Qingdao Jinrilai Graphite Co. Ltd. The granularity is 325 mesh and the content of fixed carbon is99.95 %. The following materials that are analytical pure were purchased from Sinopharm Chemical Reagent Co. Ltd.: Concentrated sulfuric acid (H_2SO_4, ≥98 %); Hydrochloric acid (HCl, 36–38 %); Potassium permanganate ($KMnO_4$, ≥99.5 %); $NaNO_3$ (99.0 %); CH_3CH_2OH (99.5 %); $NH_3{\cdot}H_2O$ (25–28 %); Sodium dodecyl benzene sulfonate (SDBS, 90.0 %); H_2O_2 (≥30.0 %); Sliced paraffin (melting temperature range is between 48 °C to 50 °C); Ethylenediamine (EDA, AP ≥99 %) was purchased from Shandong Xiya Chemical Industry Co., Ltd.

2.2. The preparation of modified graphene (E-rGO)

Graphene oxide (GO) was prepared by using the modified Hummers method [17]. 0.3 g GO was ultrasonic dispersed in 300 ml of deionized water to obtain a GO aqueous dispersion. The ultrasonic power was set to be 50 W, 90 W, 105 W and 180 W. Then the GO aqueous dispersion was transferred to a 500 ml three-necked flask. 0.9 ml ammonia was added to the flask, adjusting pH value to be 10. Stir at room temperature for 10 min, adding 180 μL ethylenediamine. Set the temperature of the water bath to be 80 °C and reflux for 24 h. Washed with anhydrous ethanol and deionized water several times and then freeze-dried for 48 h to obtain ethylenediamine synchronous reduction modified graphene (*E*-rGO) (called modified graphene (E-rGO) in the following).

2.3. The preparation of modified graphene phase change microcapsules

1 g sodium dodecyl benzene sulfonate (SDBS) was dissolved in 200 ml deionized water, stirred evenly and transferred to a 500 ml three-necked flask. 10 g sliced paraffin was added to the flask and heated to 60 °C. The solution was stirred for 3 h at the speed of 1200 rpm to obtain stable paraffin emulsion. Took 0.1 g, 0.5 g, 1 g of the modified graphene (*E*-rGO) prepared in 2.2 respectively and added them into the deionized water whose pH value is 1. Dispersed it by ultrasonic under the 100 W power for 1 h and then obtained an *E*-rGO aqueous dispersion. The E-rGO aqueous dispersion was added dropwise to the paraffin emulsion, being stirred at the speed of 1200 rpm for 2 h. The products obtained were washed with ethanol and deionized water alternately for three times, then extraction filtered and dried at 40°Cfor 24 h. 1 %, 5 %, 10 % modified graphene phase-change microcapsules were obtained, which were recorded as *E*-rGO-1/Paraffin, E-rGO-5/Paraffin, E-rGO-10/Paraffin, respectively.

Fig. 1 shows a diagram of the preparation process of the modified graphene phase change microcapsule.

2.4. Characterization method

The Sirion field emission scanning electron microscope (FEI) was used to observe the microcosmic morphology of microcapsules. The working voltage is 20.0 kV.

The phase analysis was processed on a D8 discover X-ray diffraction. Tube voltage was 40 kV and tube current was 35 mA. Scanning speed of 0.15 s/step, a step width of 0.02°and scanning range 2θ was from 5° to 70°.

The Fourier transform infrared spectra of samples was tested on a Nicolet IS10 Fourier transform infrared spectrometer of American Thermoelectric Company. The sample were mixed with KBr at a ratio of 1: 100. The wave number range was from 500 cm^{-1} to 4000 cm^{-1}.

The thermal properties were carried out by using a 200F3 DSC analyzer from the German NETZSCH company. The samples were put in N_2 atmosphere. The heating rate was 5 °C/min and the range of heating/cooling temperature ranged between −20 °C and 80 °C.

The thermal conductivity and thermal diffusivity of the samples were tested by using a Hot Disk TPS 2500S thermal constant analyzer from K-Analys AB, Sweden. The sample was made into a cylinder with a diameter of 30 mm and a height of 10 mm. Each sample was tested 3 times and an average value was obtained.

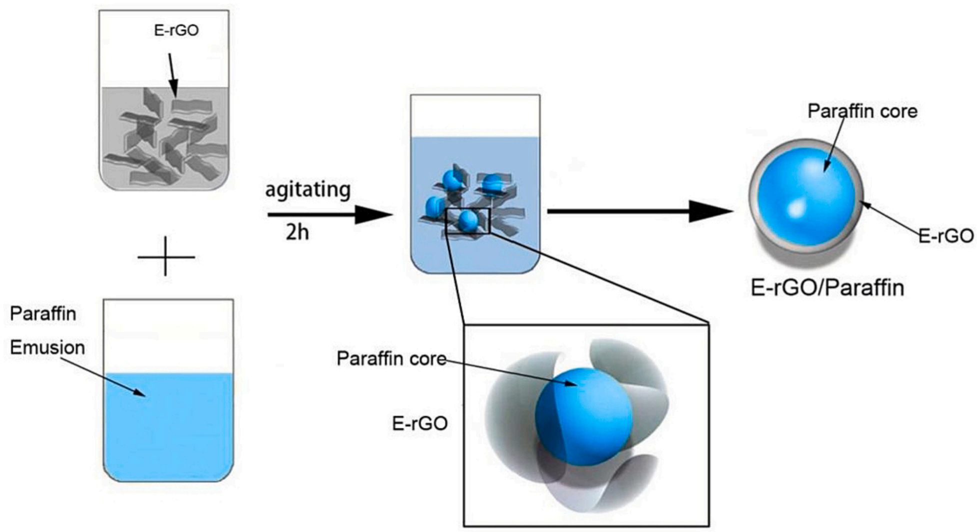


Fig. 1. The diagram of the preparation process of the modified phase change microcapsule.

3. Results and discussion

3.1. X-ray diffraction analysis

Fig. 2 shows the XRD patterns of natural flake graphite (NFG), graphene oxide (GO) and modified graphene (*E*-rGO). It can be seen that the intense and sharp diffraction peak (line (a)) of NFG is at 2θ = 26.65°, which indicates that the crystallinity of NFG is very high. This peak is formed by the stacking of (002). The interplanar spacing is calculated to be 0.33 nm based on the Bragg equation: $n\lambda = 2d\cdot\sin\theta$ ($n = 1$, $\lambda = 0.15418$ nm, $2\theta = 26.65°$). After NFG is oxidized to GO by concentrated sulfuric acid and potassium permanganate (line (b)), 2θ becomes 10.76°. The reason is that concentrated sulfuric acid and potassium permanganate introduce a large amount of oxygen-containing groups such as -OH, -COOH, -C = O and -CH (O) CH- on the upper/lower surfaces and edges of the carbocyclic ring. These oxygen-containing functional groups distract the graphite sheet and make the interlayer spacing increased, resulting in disappearance of the diffraction peak of the (002) crystal plane and generation of a new (001) crystal plane at 2θ = 10.76°. The interplanar spacing is increased to 0.823 nm by calculation. The fact demonstrates that NFG with high crystallinity has been stripped. After GO is modified by ethylenediamine (line (c)), the diffraction peak of the (001) plane disappears and the diffraction peak of the (002) plane reappears at 2θ = 23.54° with a layer spacing of 0.381 nm, which indicates that GO has been reduced to graphene. The re-emerging (002) diffraction peak gets wider and the peak intensity gets weaker compared to that of NFG. The weak peak shape still exists at 11.47°. This phenomenon may be caused by the following two reasons: First, the reducing capacity of ethylenediamine is relatively weak. After oxidation, the grafted oxygen functional groups have not been completely reduced and there are still some residual. Second, GO is grafted with -NH- while being reduced by ethylenediamine, resulting in the increase the disordered crystal plane and lower the peak intensity. In addition, a small and sharp peak appears at 29.34°. It is because the ammonia water used during the graphene reduction reacts with carbon dioxide in the air and forms a small amount of urea in the crystalline state at normal temperature, which is attached to the graphene surface.

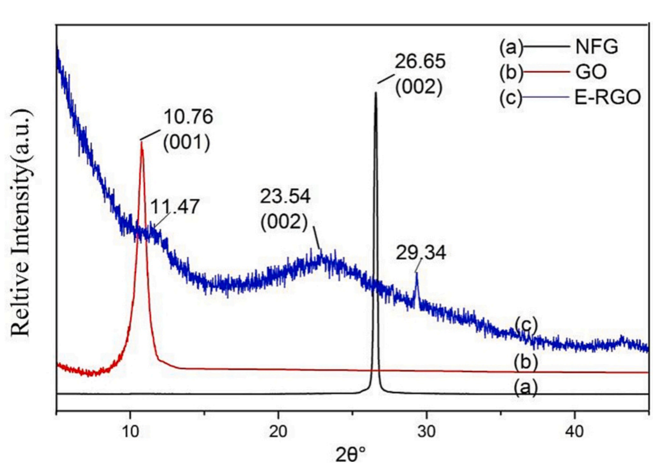


Fig. 2. XRD pattern of (a) Natural flake graphite (NFG); (b) Graphene oxide (GO); (c) Modified graphene (*E*-rGO).

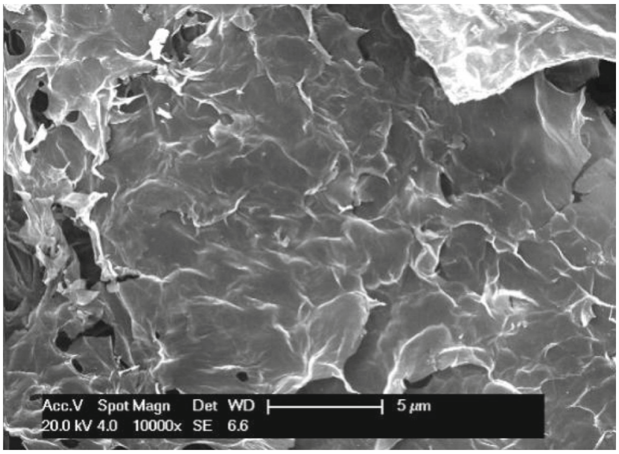
3.2. The microstructure of graphene

Fig. 3 shows the morphology of NFG, GO and E-rGO. Fig. 3 shows that the surface of NFG is smooth. The surface of GO is more dense and flat than that of NFG. A large number of oxygen-containing functional groups are inserted to the upper and lower surfaces of graphene after oxidizing, which makes the GO surface more dense. Compared to GO, the microstructure of *E*-rGO is loose and the surface is unsmooth [18]. The reason is that, after being reduced by ethylenediamine, the amine groups prevent the stacking of single graphene. Fig. 3(c) shows that a small amount of folds appear on the surface of *E*-rGO due to the high surface energy of the monolithic graphene surface, which is about 2600 m^2/g. In order to maintain a relatively stable state, the surface is wrinkled. The morphological structure of graphene varies from two-dimension to three-dimension [19]. A small number of holes appears on the surface of *E*-rGO because the ultrasound of the reduction process leads to the decrease of the size of the graphene. In addition, the grafted amine group leads to the increase of the interlayer spacing of *E*-rGO, resulting in the appearance of holes.

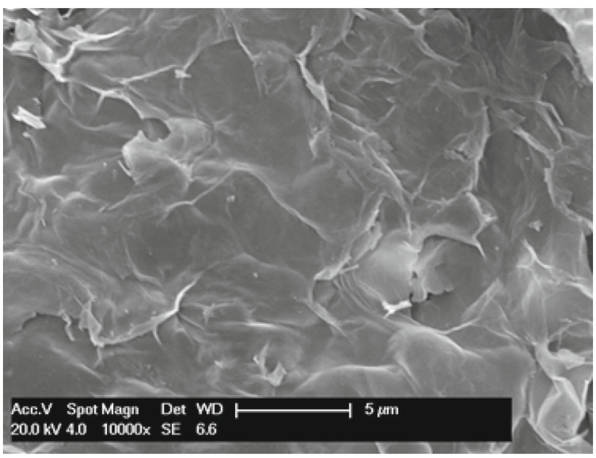
3.3. The effect of ultrasonic power on the morphology of microcapsules

Fig. 4 (a) (c) (e) (g) are SEM images of E-rGO stripped under different conditions of ultrasonic power. Fig. 4 (b), (d), (f) and (h) are SEM images of phase-change microcapsules E-rGO-10/Paraffin. It can be seen that when the ultrasonic power is 105 W, the degree of sphericity of E-rGO-10/Paraffin (Fig. 4 (f)) is high and the surface is smooth. The particle size of it is about 50 μm.

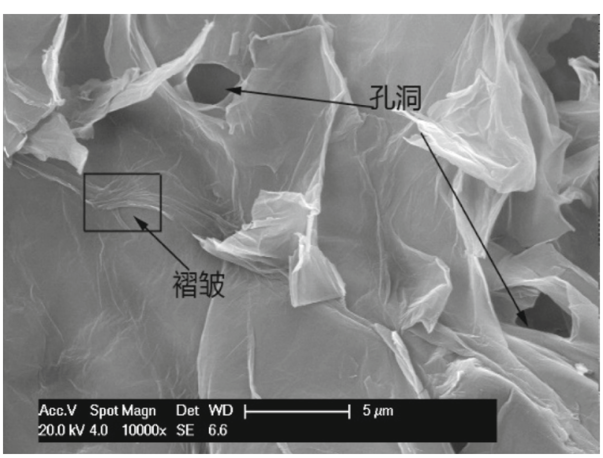
The stripping process of GO before reduction has a great influence on



(a)



(b)



(c)

Fig. 3. SEM morphology of (a) natural flake graphite (NFG); (b) GO; (c) modified graphene (*E*-rGO).

the morphology of the reduced graphene. Therefore, controlling the stripping process of GO is an important factor to fabricate the phase change microcapsules with regular morphology. Fig. 4 shows that when the ultrasonic power is 50 W (Fig. 4 (a)), the sheets of the modified graphene are not obvious, and there are some adhesions between the sheets. A few free particles still exist. When this kind of modified graphene is used as the wall material to encapsulte paraffin droplets, only a small amount of the microcapsules prepared are spherical. Most microcapsules have irregular shape (Fig. 4 (b)). When the ultrasonic power is 90 W and 105 W (Fig. 4 (c), (e)), *E*-rGO are mostly in the form of a single layer or few layers and the sheet size is large. The soft state of the single layer makes it easier to wrap paraffin droplets. The grafted amine groups on the surface of E-rGO sheets would be protonated under strong acid conditions and they are positively charged, which can bonded with the negative charges of paraffin droplets by van der Waals forces. So, the morphology of the obtained microcapsules is uniform sphere and the capsule effect is good (Fig. 4 (d), (f)). However, when the power reaches or exceeds 180 W, high-power and high-frequency oscillations would cause multi-layers oxydic graphene separating into small layers or single-layer GO. Subsequently, they will firstly break at the defect position under the action of high frequency to form fine lamellaes which are crosslink together. So the large lamellar structure would gradually changed into the porous network structure (Fig. 4 (g)). The rigidity of the network structure is large, which makes it difficult to adhere closely to the surface of the spherical paraffin droplet. So the prepared microcapsules have irregular shapes (Fig. 4 (h)).

3.4. *The infrared spectroscopy analysis*

Fig. 5 is the infrared spectra of E-rGO, Paraffin, E-rGO-10/Paraffin. As shown in Fig. 5 (a), the broad absorption peak located at 3423 cm-1 is the superimposition of the stretching vibration peak of O—H and N—H bonds. The strong peak at 1546 cm-1 is the characteristic absorption peak of N—H in-plane bending vibration, while the peak at 678 cm-1 is the characteristic absorption peak of N—H out-of-plane bending vibration. These two peaks demonstrated the existence of amine groups in E-rGO. The peak at 1643 cm-1 is the stretching vibration absorption peak of –C=C-. The symmetrical bending vibration absorption peak of -CH3 locates at 1374 cm-1. The peak at 1167 cm-1 and 803 cm-1 is the vibration absorption peak of C—O. This is due to the limited reducing ability of ethylenediamine. It is difficult to remove all the oxygen-containing functional groups grafted on the surface of graphene. Combined with the analysis of XRD patterns in Fig. 2, it can be concluded that the amine group of ethylenediamine reacts with hydroxyls or epoxy groups located on the same side of GO sheets to perform cyclization, followed by removal reaction [20].

Combined with the analysis of XRD patterns in Fig. 2, it can be concluded that the amine group of ethylenediamine reduces and modifies the GO by reacting with the oxygen-containing groups on GO in the way of substitution reaction. The substitution reaction cannot remove completely the oxygen-containing functional group, so *E*-rGO has a graphite-like structure, which is not exactly the same as graphite structure [21].

In Fig. 5 (b), the peak at 3604 cm^{-1} is the free -OH characteristic absorption peak. The broad peak at 3429 cm^{-1} corresponds with the characteristic absorption peak of the association state -OH. The peaks at 2965 cm^{-1} and 2639 cm^{-1} are the C—H in -CH_3 asymmetric stretching vibration absorption peak. The peak at 2848 cm^{-1} is the stretching vibration peak of -CH2 and the peak of 1374 cm^{-1} is the symmetrical bending vibration absorption peak of C—H in -CH_3. The peaks at 1894 cm^{-1}, 1467 cm^{-1},722 cm^{-1} correspond the in-plane bending vibration peak, bending vibration absorption peak and in-plane swing bending absorption peak of -CH_2 -, respectively. The intensity of the peak at 1123 cm^{-1} is weak, which is the symmetrical vibration absorption peak of -C-O-C. Because there are a small amount of olefins that exist in the paraffin, the peaks at 2339 cm^{-1}, 2025 cm^{-1} and 1624 cm^{-1} are the

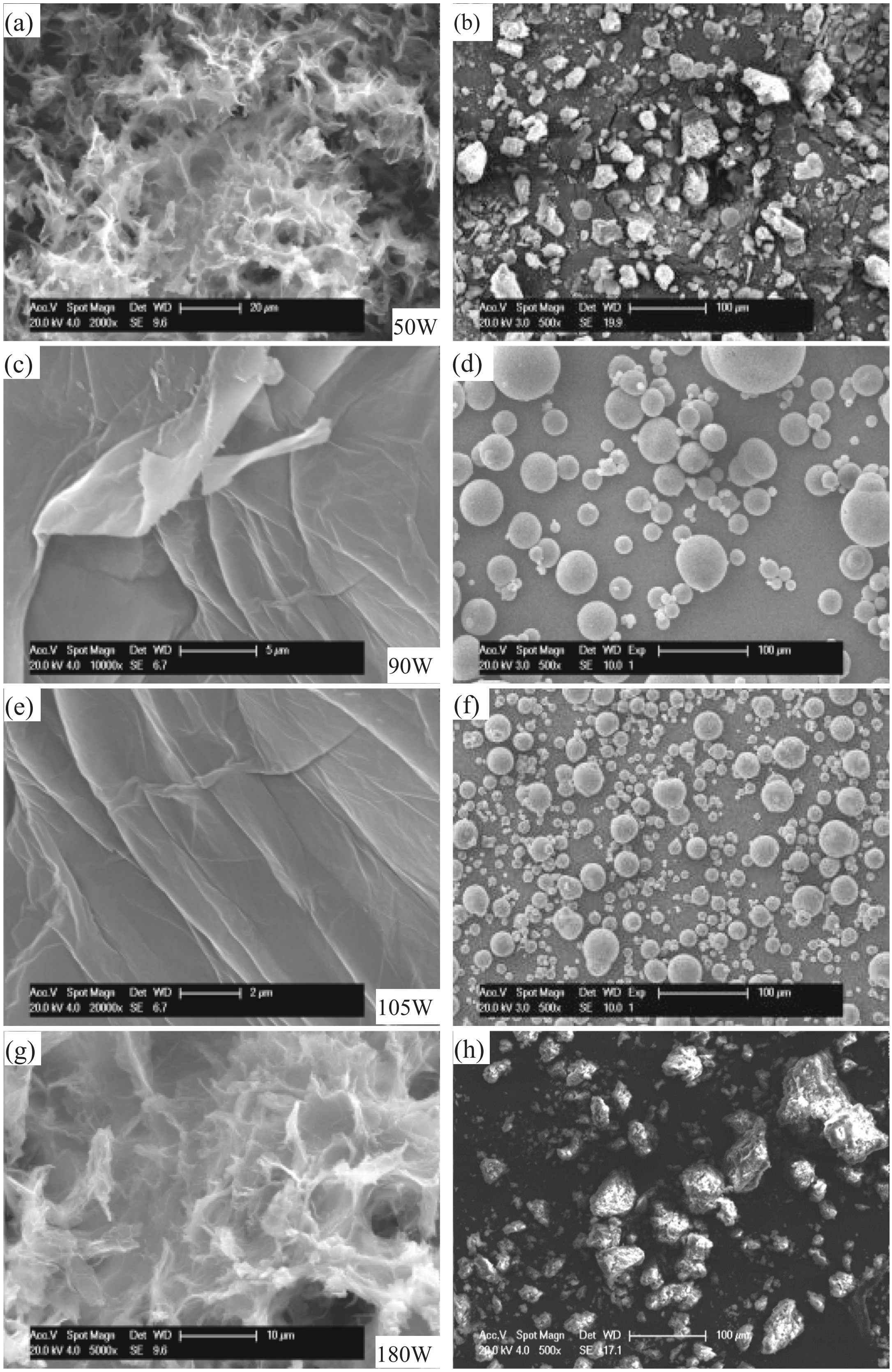


Fig. 4. The SEM morphology of *E*-rGO and E-rGO-10/Paraffin under different ultrasonic powers.

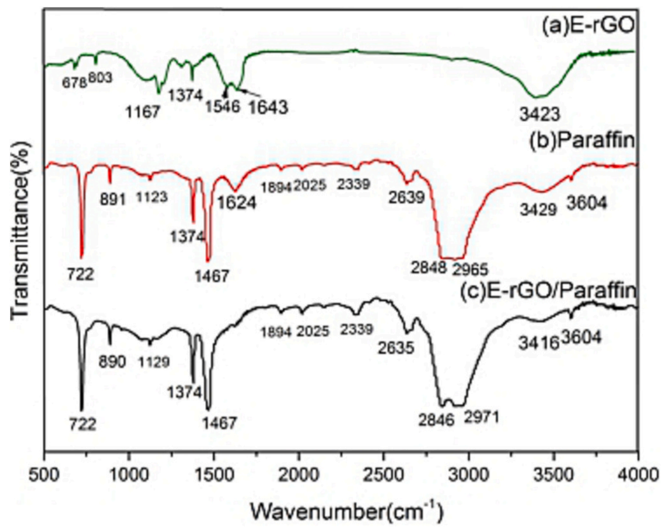


Fig. 5. The infrared spectra of E-rGO, Paraffin and E-rGO-10/Paraffin.

stretching vibration absorption peak of –C=C- and the peak at 891 cm^{-1} is the bending vibration absorption peak of = C-H.

Through comparing Fig. 5(a), (b), (c), it can be seen that the intensity of the peak at 1467 cm^{-1} is strong. It overlaps with the characteristic absorption peak of N—H in-plane bending vibration at 1546 cm^{-1} in Fig. (a), which makes the N—H absorption peak less visible. Besides, after paraffin is coated by *E*-rGO, the position of the characteristic peak basically remained unchanged, but there was a small amount of offset and the offset was within the range of each characteristic peak frequency. This is because although the frequency of a group is mainly determined by the atoms in the group itself, variations in the internal structure of the molecule and the external environment would also affect the group frequency [22]. The fact that offsets are within the range of the characteristic peak frequency shows no new functional groups appear in the process of the formation of microcapsules. The encapsulation of paraffin with wall materials is only a physical effect, and there is no generation of new substances. Therefore, the use of E-rGO to wrap paraffin does not change the structure and properties of paraffin itself.

3.5. Thermal performance analysis

Fig. 6 shows the DSC curves of phase change microcapsules with the content of 5 % and 10 % *E*-rGO, respectively. The thermal storage performance parameters are listed in Table 1, ΔH_m is the enthalpy of melting and ΔH_s is the enthalpy of solidification; T$_m$ is the initial melting temperature and T$_s$ is the initial solidification temperature; Pm is the peak temperature of melting and Ps is the peak temperature of freezing. The coverage of the microcapsules (R) is calculated according to Eq. (1) [23]:

$$R = \frac{\Delta H_{m,MicPCM}}{\Delta H_{m,PCM}} \times 100\% \tag{1}$$

Among them: $\Delta H_{m,MicPCM}$ is the phase change enthalpy of phase change microcapsules, $\Delta H_{m,PCM}$ is the phase change enthalpy of phase change microcapsules.

As shown in Fig. 6 and Table 1, microcapsules have obvious absorption and exothermic peak, indicating that they have a certain energy storage capacity. The initial melting temperature and the peak temperature of *E*-rGO-5 and E-rGO-10 are both decreased compared with those of paraffin. It is because there is an interfacial effect between graphene and paraffin. In other words, when some segments of the paraffin chain extend to the interface between the paraffin and the graphene, these chains in the interface have more free space for the slippage movement and so the movement of the paraffin segment is

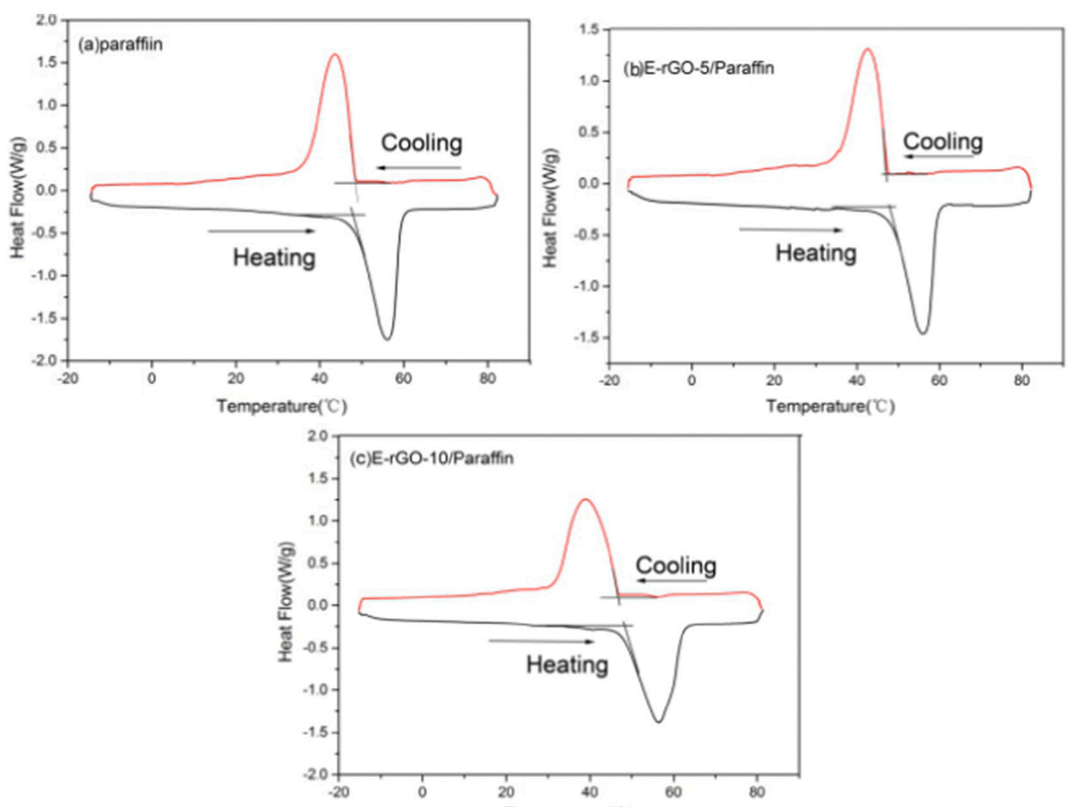


Fig. 6. The DSC curves of paraffin and phase change microcapsules with different graphene content.

Table 1
The thermal storage performance parameters of paraffin and phase change microcapsules with different graphene content.

Samples	ΔH_m (J/g)	ΔH_s (J/g)	T_m (°C)	T_s (°C)	P_m (°C)	P_s (°C)	Encapsulation Ratio(%)
Paraffin	112	117.4	49	48.6	56.3	43.7	–
E-rGO-5/Paraffin	94.1	95.3	48.2	47.6	55.8	42.2	84.02
E-rGO-10/Paraffin	100.3	98.85	47.8	47.6	55.3	39.4	89.5

increased. As a result, the phase transition temperature decreases after paraffin is encapsulated by the wall material to form the microcapsules.

As shown in Table 1, the enthalpy of phase change of *E*-rGO-5 and E-rGO-10 is lower than that of paraffin. This is because the energy density of the microcapsules after encapsulation is lower than that of paraffin. Table 1 shows that, with the increase of the modified graphene content, the encapsulation ratio of the phase-change microcapsules increases.

3.6. Thermal cycle stability

After experiencing different times of heating/cooling cycles, the DSC curves of E-rGO-10/Paraffin microcapsules prepared under 90 W power are shown in Fig. 7. As shown in Fig. 7, the trends of the DSC curves before and after heating/cooling cycles are basically the same. The peak area of absorption and exothermic after the heating/cooling cycles decreased slightly. Table 2 shows the phase change of enthalpy (ΔH_m), the decrease value of enthalpy (ΔH_m), the peak temperature of melting (P_m) and the initial melting temperature (T_m) of E-rGO-10/Paraffin after heating/cooling cycles. It can be seen from Table 2 that after 50 cycles and 100 cycles the peak temperature of melting decreases by 0.1 °C and 0.3 °C, respectively. The initial melting temperature (T_m) increases by 0.2 °C and 0.4 °C, respectively. The result indicates that E-rGO-10/Paraffin can maintain a relatively stable phase change temperature after repeated many times of cycles. After 50 cycles, the enthalpy of phase change of the microcapsule decreases by 3.61 %. After 100 cycles, the enthalpy of phase change of the microcapsule decreases by 6.51 %. It means that a small amount of paraffin leaks after many times of heating-cooling cycles. This is because the graphene oxide aqueous dispersion has been ultrasonically stripped before reducing the graphene oxide, so the size of the graphene sheet decreases and the interlayer spacing increases, resulting in gaps in the formed wall materials. The leakage quantity is still within the acceptable range, which does not influence the application of the prepared microcapsules. Silakhori et.al [24] found that the latent heat of microencapsule has a variation of 3.3–5.8 % for the freezing process after 200 to 1000 cycles compared with zero cycles.

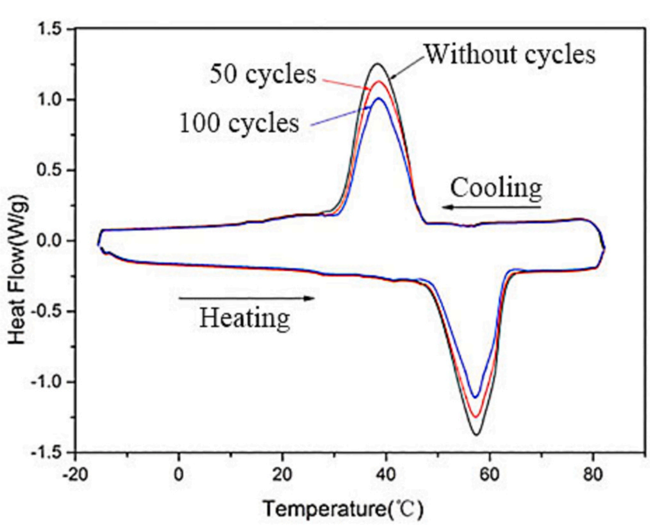


Fig. 7. The DSC curves of E-rGO-10/Paraffin: without cycles, after 50 cycles, after 100 cycles.

Table 2
The thermal storage performance parameters of E-rGO-10/Paraffin after heating/cooling cycles.

Times of cycles	ΔH_m(J/g)	The decrease value of ΔH_m (%)	P_m(°C)	T_m(°C)
0	100.3	0	55.3	47.2
50	96.68	3.61	55.2	47.4
100	93.77	6.51	55.0	47.6

Thermal cycling tests were performed by Shukla et.al to check the stability of phase change materials [25]. The results showed that the melting temperature decreased by 31 °C and latent heat decreased by 28 kJ/kg. In this paper, the enthalpy of phase change of the microcapsule decreases by 6.51 %, which is similar with the above results. There is no significant change in the latent heat capacity.

3.7. Thermal conductivity

Fig. 8 shows the thermal conductivity of Paraffin, E-rGO-1/Paraffin, E-rGO-5/Paraffin and E-rGO-10/Paraffin, respectively. As shown in the figure, the thermal conductivity of the phase change microcapsules increases after using the modified graphene to encapsulte paraffin. The thermal conductivity of the microcapsules gradually increased with the increase of the content of E-rGO.

The thermal performance parameters of phase change microcapsules with different content of graphene are shown in Table 3. It can be seen that the thermal conductivity of phase change microcapsule is 0.264 W/(m· K) when the content of modified graphene is 1 %, which is 1.39 times higher than paraffin. When the content of modified graphene is 5 %, the thermal conductivity increases by 1.78 times. When the content of modified graphene is up to 10 %, the thermal conductivity increases to 4.01 times. However, compared with the extremely high thermal conductivity of graphene, the increase of the thermal conductivity of graphene phase change microcapsules is not so high. This is because the surface of graphene obtained by oxidation-reduction has folds and there

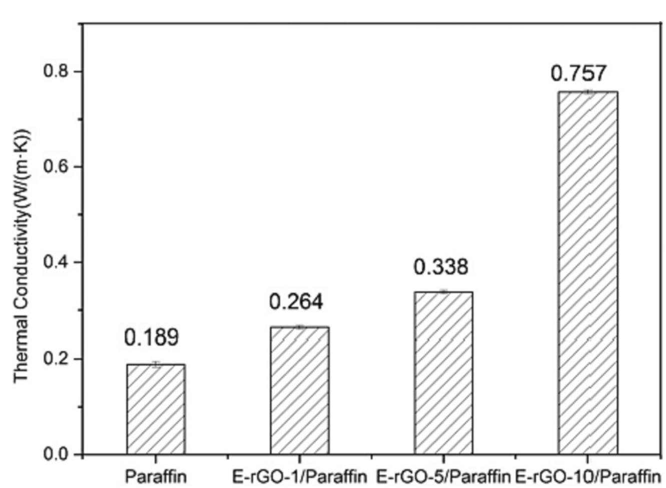


Fig. 8. Thermal conductivity of phase change microcapsules with different content of graphene.

Table 3
The thermal performance parameters of phase change microcapsules with different content of graphene.

Parameters	Paraffin	E-rGO-1/ Paraffin	E-rGO-5/ Paraffin	E-rGO-10/ Paraffin
Thermal conductivity (W/(m·K))	0.189	0.264	0.338	0.757
Thermal diffusivity (m^2/s)	0.09	0.142	0.191	0.428
Specific heat capacity (J/(g·K))	2.103	2.071	1.773	1.963

are gaps between the layers, which leads to the fact that the thermal resistance of the interface increases when heat is transferred between the graphene sheets. In addition, there is interface thermal resistance exist between graphene and paraffin. So the increase of graphene microcapsules thermal conductivity is limited by these factors.

4. Conclusions

In this paper, ethylenediamine was used to reduce graphene oxide and aimino-group grafted onto the surface of graphene. The thermally conductivity enhanced phase change microcapsule was prepared by using the modified graphene as the wall material and the paraffin as the core material. Through the analysis of the microstructure, heat storage performance, thermal cycle stability and thermal conductivity, the following conclusions are drawn:

(1) The microcapsules prepared under 105 W power of ultrasonic stripping have regular spherical shape with an average particle size of 50 μm.
(2) With the increase of the content of the modified graphene, the thermal conductivity of the microcapsules increases gradually. When the content of graphene is 10 %, the thermal conductivity of the microcapsules increases to 0.757 W/ (m·K), which is 4.01 times that of paraffin. The wrapping rate is up to 89.5 % and the enthalpy of phase transition of the microcapsules is 100.3 J/g.
(3) Using the modified graphene to wrap paraffin does not cause any chemical changes and does not change the energy storage properties of the paraffin.
(4) The modified graphene phase change microcapsule is a kind of energy storage material with high thermal conductivity, strong energy storage capacity and good thermal cycle stability.

CRediT authorship contribution statement

Min Li: Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Qiuge Ma:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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